Multiobjective Optimization Paradigm for Toroidal Inductors With Spatially Tuned Permeability

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***Abstract*—Spatially tuning core permeability of an electromagnetic device enables superior performance. A permeability profile can be heuristically selected to improve the flux distribution in a device with a given geometry, but in order to fully leverage the capacity of spatial dependent permeability engineering, the geometry and the permeability should be optimized simultaneously. The work in this article herein presented sets forth a multiphysics design optimization paradigm that includes the permeability profile tuning in the context of both inductor and converter design. This approach enables the determination of Pareto optimal fronts consisting of a set of optimal solutions against competing objectives (e.g., mass and loss) under imposed constraints. To this end, computationally efficient analytical solutions of the heat transfer and electromagnetic formulations are derived for toroidal inductors, which are validated with finite-element analysis-based simulations. The software implemented in MATLAB 2018b is available online as an attachment to this article.**

***Index Terms*—Design optimization, inductor, multiphysics, permeability tuning, spatial permeability profile, toroid.**

I. Introduction

POWER conversion equipment utilizes energy storage devices to manage power flow during switching cycles. These passive components constitute a substantial portion of the volume and mass of typical power converters [1], [2]. Therefore, an important step in the miniaturization of these systems is the design optimization of passive devices, including magnetic components [3], [4]. This process commonly involves the selection of an appropriate geometry with the corresponding optimal dimensions, winding configuration, and a core material [5]. However, uneven distribution of flux density makes certain areas of the core more saturated than others. The utilization factor of the magnetic core can be maximized by the spatial manipulation of the permeability across the device. Therefore, permeability engineering as part of the device design process leads to greater converter power density.

A certain degree of permeability engineering can be achieved by layering magnetic cores of different permeabilities to achieve a more uniform overall flux density distribution. Designated as multipermeability inductors, this approach relies on a manufacturable discretization of an idealized variation of the permeability across the magnetic core [6]–[12]. In [6], this concept is investigated using ferrite tapes to create layers of different permeability. An idealized linear profile of the permeability is defined, and subsequently discretized into a number of sections for manufacturing. The staircase approximation of the permeability profile that best approximates the desired spatial dependent material property is built with layers of permeabilities available of each ferrite tape. The inductor geometries investigated are single turn with circular and square cross sections of the core, where the permeability is varied radially away from the inner conductor. From a modeling perspective, the permeability is assumed to be linear until saturation, after which it is assumed to be μ_0. This model is improved into a piecewise linear approximation of the BH curve in [7]. Note that as the core is discretized into thinner layers of different magnetic properties, the permeability profile tends to a continuous description. However, the designer is limited to a finite set of permeabilities available for each layer. In [9] and [12], the multipermeability inductor is built by axially stacking layers, and a greater inductance is enabled by a series connected multiwindow configuration distributed in an array around the cross section of the core. The concept of a multipermeability toroidal inductor is implemented using additive manufacturing in [11]. A toroid is extruded from three magnetic pastes (sinterable ferrite paste and curable powdered core), each with a different permeability, forming three concentric rings for a better distribution of flux density.

Recent developments in magnetic material treatment have enabled a continuous spatial permeability tuning around the core [13]–[15]. In [15], the concept employed in [11] is taken one step further to obtain a continuous permeability profile around a toroid through gradient sintering. By leveraging the linear relationship of the sintering temperature of a ferrite with its resultant permeability, a continuous spatial tuning can be

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achieved [16]. That process requires only one material, NiZnCu ferrite, and a controlled temperature gradient across the core to obtain a permeability profile in a toroidal inductor. In [13], the spatial manipulation of permeability in metal amorphous nanocomposite (MANC) alloy cores is reported. These materials are commonly referred as nanocrystalline core materials and some chemistries are commercially available (e.g., FINEMET). The performance of several applications in power electronics are enhanced using these materials [17]–[21]. Also, the object of study in this article, MANC alloys are a soft magnetic material with properties determined by its composition optimization and controlled annealing treatments, which produce a nanocomposite structure of nanocrystals in an amorphous precursor material. These alloys are produced through a rapid solidification process to form long amorphous metal ribbons (AMRs) of approximately 13–25 μm in thickness. The AMRs (amorphous phases) are exposed to annealing treatment to obtain nanocrystalline phases, creating the MANC nanocomposite structure. Therefore, during manufacturing, nanocrystalline cores require an annealing step, which can be done after winding the metal ribbon into the core final shape (core annealing) [22], before (ribbon annealing) [13], [23], or both before and after [24]. Annealing the ribbon before winding the core can be achieved through roll-to-roll in-line annealing. Compared to postwinding annealing, the in-line process facilitates temperature control of the tensioned metal ribbon, which is challenging to control in large core stacks because of ribbon self-heating due to latent crystallization heat [13], [24]. Furthermore, using the appropriate alloys, in-line annealing treatments with a controlled mechanical tension applied to the metal ribbon (strain annealing) can be applied to influence the magnetic properties of the final product [25]. The treated magnetic ribbon is subsequently wound into its final geometry. By changing the initial composition of the AMR and the applied tension during processing, relative permeabilities from as low as $\sim$10 up to as high as $\sim 10^4$ can be obtained [25]. Furthermore, as the treated MANC ribbon is wound in its final shape, at under 25 μm thick, a virtually continuous description of the permeability spatial profile can be realized for power magnetic devices. A permeability profile can be selected to improve the performance of a device with a given geometry, but to fully leverage the capacity of spatial dependent permeability engineering, the geometry and the permeability should be optimized simultaneously. Therefore, an effort has been set up to set forth an optimization paradigm that not only yields an optimal inductor design for a given set of requirements and permeability, but also determines the desired spatial variation of permeability as part of the design process. The result of this work enables the designer to inform the core manufacturer about the optimal permeability profile for their application. This profile can be correlated to the appropriate alloy and tension control during in-line strain annealing by a core manufacturer.

The work in this article considers a toroidal geometry. It consists of the toroidal core of rectangular cross section wrapped by a protective layer, and then the coil around it. Depending on the number of layers of the winding, due to the greater circumference on the outer radius, the inner region of the coil may have more layers than the outer region.

The feasible solution space of an electromagnetic device multiobjective optimization is often discontinuous and nonconvex; hence, evolutionary algorithms are typically employed to solve these problems. Therefore, in order to produce a Pareto optimal front, numerous design evaluations are required, which makes advantageous the use of computationally efficient models. Therefore, the models herein developed are semianalytical descriptions derived from the governing physics. Hence, the optimization relies on the analytical solution to determine the electromagnetic performance of the inductor, including magnetizing and leakage flux. In inductor optimization, the device miniaturization and material savings are often limited not by the electromagnetic performance, but rather by thermal limits. Thermal equivalent circuits (TECs) that capture the anisotropy of thermal conductivities of the core and coil are included in the optimization.

The contribution of this article is twofold: the evaluation of spatial permeability engineering in contextual multiobjective optimization and the introduction of a computationally efficient multiphysics model for toroidal inductors design. Note that the model introduced herein can, be applied to design inductors using any magnetic core, not only those that enable spatial permeability optimization. Furthermore, a computationally efficient inductor design model can be introduced in an overarching power electronic converter optimization problem. In such paradigm, the converter components and operational parameters are simultaneously optimized with the inductor geometry and permeability profile.

The rest of this article is organized as follows. Section II describes permeability function characterization. Section III introduces the geometry function. Section IV presents the electromagnetic model and Section V the heat transfer model. Section VI provides results and insights about permeability optimization in an ac inductor design, and a buck converter optimization. Finally, Section VII concludes this article.

II. Material Property Characterization

Spatial permeability profiling offers significant benefits to the inductor design. Rather than assuming that any magnetic material can be arbitrarily manipulated, this work considers experimentally determined achievable properties of strain annealed MANC in the optimization paradigm. Therefore, a permeability description function is modified to be tuned and fitted to experimentally observed data. That is enabled from the definition of flux density B as

$$B \triangleq \mu_0 H + M \tag{1}$$

where μ_0 is the magnetic permeability of free space, H is the field intensity, and M is the magnetization. The magnetic permeability of the core, defined as the ratio B/H, can be modeled as a function of B by rearranging (1) as follows:

$$\mu(B) = \mu_0 \frac{\Gamma(B)}{\Gamma(B) - 1} \tag{2}$$

where the function $\Gamma(B)$ is the ratio B/M as defined in [26]. Note that setting up the permeability tuning as a proportionality

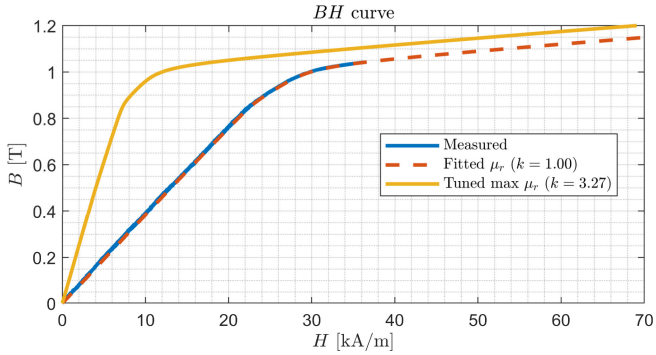


Fig. 1. Measured, fitted, and tuned BH curves for strain annealed $Co_{74.6}Fe_{2.7}Mn_{2.7}Nb_4Si_2B_{14}$ using the parameters shown in Table I.

TABLE I
STRAIN ANNEALED $Co_{74.6}Fe_{2.7}Mn_{2.7}Nb_4Si_2B_{14}$ BASE PERMEABILITY CURVE

Parameter	Permeability Material Properties			
μ_r	30.5658			
i	1	2	3	4
α_i	0.3396	0.023712	0.0063699	0.0015238
β_i	56.03812	61.81279	26.04038	1000
γ_i	1.0284	0.86461	1.1314	0.46935

factor, i.e., $k\mu$(B), would result in a nonphysical permeability in saturation ($k\mu_0$). Therefore, consider expressing the magnetic permeability as in (2), introducing the tuning factor k in the ratio function $\Gamma(B)$ as

$$\Gamma(B,k) = \frac{k\mu_r}{k\mu_r - 1} + \sum_{i=1}^{N_t} \alpha_i |B| + \delta_i \ln\left(\varepsilon_i + \zeta_i e^{-\beta_i |B|}\right) \tag{3}$$

where k is the tuning factor, μ_r is the relative magnetic permeability, N_t is the number of terms used to curve fit the material properties, and the remaining factors are

$$\delta_i = \frac{\alpha_i}{\beta_i}; \quad \varepsilon_i = \frac{e^{-\beta_i\gamma_i}}{1 + e^{-\beta_i\gamma_i}}; \quad \zeta_i = \frac{1}{1 + e^{-\beta_i\gamma_i}} \tag{4}$$

where α_i, β_i, and γ_i are fitted parameters. Appropriately, $\mu \to \mu_0$ as $B \to \infty$. From the measured material data, the anhysteretic characteristic of the loop is extracted by averaging the data upper and lower curves. Then, the parameters in (4) are fitted to experimentally measured data, as shown in Fig. 1. The fitted parameters for the sample of $Co_{74.6}Fe_{2.7}Mn_{2.7}Nb_4Si_2B_{14}$ strain annealed to the permeability $\mu_r \approx 30$ are shown in Table I. The strain annealed minimum and maximum permeability for this alloy is $\mu_r \approx 10$ and $\mu_r \approx 100$, respectively, hence the optimization that will be herein discussed allows the tuning factor to range k from 1/3 to 3.27. Table II summarizes further parameters used for the optimization, i.e., mass density (ρ), Steinmetz loss parameters (α, β, k_h), and thermal conductivities in the radial (k_r) and axial (k_z) directions.

TABLE II
STRAIN ANNEALED $Co_{74.6}Fe_{2.7}Mn_{2.7}Nb_4Si_2B_{14}$ MATERIAL PROPERTIES

Parameter	Material Properties	Steinmetz Parameter	Material Properties
k_r [W/(m·K)]	1.5	α	1.081023
k_z [W/(m·K)]	7.0	β	2.015963
ρ [kg/m^3]	5500	k_h [J/m^3]	45.61684

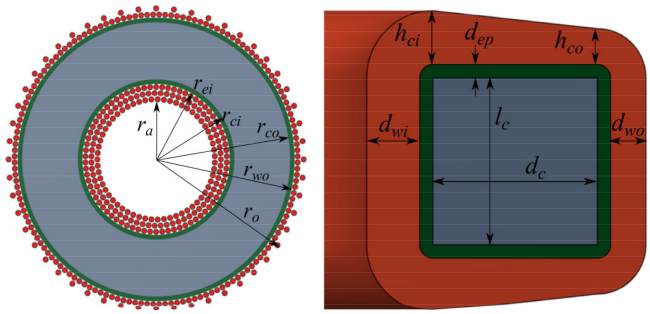


Fig. 2. Toroid geometry: top view (left) and side cross-sectional view (right).

III. GEOMETRY

The toroidal inductor geometry used for design is shown in Fig. 2. The geometry function is formulated such that dependent variables are computed from the fewest possible independent variables. To avoid geometry constraints, the mathematical geometric construction is built inside outward and it is inherently self-consistent, based from these variables. The set of independent geometry variables are

$$\mathbf{G}_I = \{N, a_c^*, r_a, l_c, d_c\} \tag{5}$$

where N is the number of coil turns, a_c^* is the desired conductor area, r_a is the finished inner radius, l_c is the core axial length, and d_c is the core radial depth, as illustrated in Fig. 2. In addition to the independent geometry variables $\mathbf{G}_I$ in (5), user-defined independent design parameters $\mathbf{D}_I$ are fixed values that are not optimized, but defined by the following set:

$$\mathbf{D}_I = \{d_{ep}, n_{\text{spc}}, k_b\} \tag{6}$$

where d_{ep} is the protection layer depth between the core and coil, n_{spc} is the number of strands per conductor, and k_b is the built factor of the winding machine. Note that the desired conductor area a_c^* is divided by the number of strands per conductor n_{spc}, and approximated to the closest allowed value in the AWG standard for wire selection. From that wire selection, the strand radius r_s and conductor equivalent radius r_c are readily available. From (5) and (6), the geometry in Fig. 2 is fully defined. In order to determine the radial winding depths d_{wi} and d_{wo}, the number of winding layers in each side needs to be computed by accumulating however many layers are needed for all strands. The ith layer hosts $N_{\text{cap}}^{(i)}$ strands as

$$N_{\text{cap}}^{(i)} = \pi / \sin^{-1}\left(\frac{r_s k_b}{r_a + (2i-1)\, r_s k_b}\right). \tag{7}$$

The same equation holds for the outer layers by replacing r_a with r_{wo}. The inner coil radial depth is then computed from the number of inner layers N_{li} by

$$d_{wi} = 2r_c k_b N_{li}. \tag{8}$$

The coil height above the inner radius of the core h_{ci} is computed by

$$h_{ci} = 2r_s k_b \text{ceil}\left(\frac{N}{\text{floor}\left(\left(2\pi r_{ci}\right)/\left(2r_s k_b\right)\right)}\right). \tag{9}$$

The coil height above and below the outer radius of the core is computed similarly, and in between that and the inner core radius, the height is assumed to vary linearly.

IV. Electromagnetic Model

In a design model for evolutionary optimization, computational efficiency is imperative. Therefore, an analytical model that captures the ac and dc electromagnetic performance is set forth in this section.

A. Magnetizing Inductance

In the toroid, the magnetizing fields can be approximated to be all oriented in the azimuth orientation, reducing the constitutive relations of the material to a scalar equation. The magnetic flux density in a point in the core is obtained by determining the root *B*∗ of the residual function *F(B)*

$$F\left(B\right) = B - \mu\left(B,k\right)H. \tag{10}$$

Combining (2), (10), and using Ampere's law

$$F\left(B\right) = B - \mu_0 \frac{\Gamma\left(B,k\right)}{\Gamma\left(B,k\right)-1}\frac{NI_c}{2\pi r} \tag{11}$$

where I_c is the current applied to the coil and *r* is the radius in cylindrical coordinates from the origin at the center of the toroid. For solving (11), its derivative with respect to *B* is

$$\frac{dF}{dB} = 1 + \left[\text{sgn}\left(B\right)\frac{\mu_0}{\left(\Gamma\left(B,k\right)-1\right)^2}\sum_{i=1}^{N_t}\frac{\alpha_i e^{-\beta_i\gamma_i}}{e^{-\beta_i\gamma_i}+e^{-\beta_i|B|}}\right] \times \frac{NI_c}{2\pi r}. \tag{12}$$

From a vector of N_p radii ranging from r_{ci} to r_{co}, a vector of N_p flux densities is obtained by using (11) and (12) in conjunction with Newton's method. This flux density is numerically integrated with respect to radius to determine the magnetizing flux linkage through the core in accordance with

$$\lambda_m\left(I_c\right) = \frac{Nl_c}{2}\sum_{n=1}^{N_p-1}\left(r_{n+1}-r_n\right)\left(B_{n+1}\left(I_c\right)+B_n\left(I_c\right)\right) \tag{13}$$

where $r_{ci} = r_1 < \ldots < r_{Np} = r_{co}$. The incremental magnetizing inductance is readily computed from a small disturbance in the current ΔI in (13). Finally, the permeability tuning factor *k* is assumed to vary with function radius as

$$k = \text{f}_\mu\left(r\right) \tag{14}$$

where $\text{f}_\mu(r)$ is an arbitrary permeability profile. In this article, a piecewise Hermite cubic polynomial is used to establish a smooth profile to facilitate manufacturing, while at the same time preventing interpolated values from going beyond the achievable permeability range.

B. Coil Leakage Permeance

The permeance related to the leakage of the winding can be derived from the energy stored in the region of the winding, which is magnetic linear, and is related by

$$E = PN^2 i^2/2 \tag{15}$$

where *P* is the permeance associated to the energy stored *E*, *N* is the number of turns, and *i* is the current. It is shown in [27] that the energy in a magnetically linear region is also expressed as

$$E = \frac{1}{2}\mu\int_V H^2 dV. \tag{16}$$

Equating (15) and (16) allows the determination of the leakage permeance in a given region of the winding. Consider first the inner side of the winding, where the conductors are vertically oriented. From Ampere's law, the magnitude of the field intensity as a function of the radius is given by

$$|\mathbf{H}| = \begin{cases} 0, & 0 \le r \le r_a \\ \frac{Ni(r-r_a)}{2\pi r d_{wi}}, & r_a \le r \le r_{ei} \\ Ni/2\pi r, & r_{ei} \le r \le r_{ci}. \end{cases} \tag{17}$$

Substituting (17) into (16), and comparing the result with (15), enables the determination of the permeance associated to the inner vertical region of the coil P_{iv}, which is the sum of the coil interior component P_{ivi} defined in $r_a \le r \le r_{ei}$, and P_{ive} defined in $r_{ei} \le r \le r_{ci}$. The expressions for each are

$$P_{ivi} = \frac{\mu_0 l_c}{2\pi d_{wi}^2}\left[\frac{d_{wi}\left(r_{ei}+r_a\right)}{2} - 2r_a d_{wi} + r_a^2 \ln\left(\frac{r_{ei}}{r_a}\right)\right] \tag{18}$$

$$P_{ive} = \mu_0 l_c \ln\left(r_{ci}/r_{ei}\right)/2\pi. \tag{19}$$

Repeating the same procedure to the outer vertical region of the coil, $P_{ov} = P_{ovi} + P_{ove}$ yields

$$P_{ovi} = \frac{\mu_0 l_c}{2\pi d_{wo}^2}\left[\frac{d_{wo}\left(r_{co}+r_{wo}\right)}{2} - 2r_o d_{wo} + r_o^2 \ln\left(\frac{r_o}{r_{wo}}\right)\right] \tag{20}$$

$$P_{ove} = \mu_0 l_c \ln\left(r_{wo}/r_{co}\right)/2\pi. \tag{21}$$

The horizontal region of the winding P_h (top or bottom) can be computed using the same approach; however, the energy integral in (16) is integrated from r_{ci} to r_{co}, to the height of the coil which is assumed to vary linearly from h_{ri} to h_{ro}, as shown in Fig. 2. The permeance component associated to the region between the core and the coil (the protection layer) is given by

$$P_{he} = \mu_0 d_{ep} \ln\left(r_{co}/r_{ci}\right)/2\pi. \tag{22}$$

The coil permeances are used to compute both the leakage flux linkage and proximity effect losses. When coupling the electromagnetic losses to the heat transfer model, it will be

TABLE III
INDUCTOR PARAMETERS FOR MODEL VALIDATION

Parameter	Value	Parameter	Value
N	150	d_c	20 [mm]
a_c^*	2.082 [mm²]	d_{ep}	0.8 [mm]
r_a	11.71 [mm]	n_{spc}	1
l_c	5 [mm]	k_b	1.05

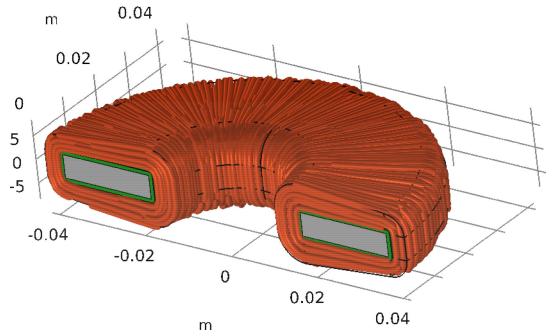


Fig. 3. Cross-sectional view of the inductor described by Table III.

useful to divide the top and bottom horizontal coil domains into N_s sections. Therefore, consider defining the radii breakpoints $r_{ci} = r_{cs,1} < r_{cs,2} < \ldots < r_{cs,Ns+1} = r_{co}$, and the coil height above each radius as $h_{ci} = h_{cs,1} < h_{cs,2} < \ldots < h_{cs,N_s+1} = h_{co}$, thus, each section permeance $P_{h,i}$ is given by

$$P_{h,i} = \frac{\mu_0}{6\pi}\left[h_{cs,i+1} - h_{cs,i} + \frac{h_{cs,i}r_{cs,i+1} - h_{cs,i+1}r_{cs,i}}{r_{cs,i+1} - r_{cs,i}} \times \ln\left(\frac{r_{cs,i+1}}{r_{cs,i}}\right)\right]. \quad (23)$$

Each horizontal coil region (top and bottom) permeance is computed by adding up the permeances for each section in (23). Subsequently, summing all leakage permeances for the inner and outer vertical sections, in addition to the horizontal regions $(P_{iv} + P_{ov} + 2P_h)$, and neglecting the corner regions of the winding, yields the total leakage permeance of the winding P_{lk}. Therefore, the leakage flux linkage can be computed by

$$\lambda_{lk} = N^2\left(P_{iv} + P_{ov} + 2\sum_{i=1}^{N_s} P_{h,i}\right) I_c. \quad (24)$$

Consider for validation, an arbitrary inductor with parameters shown in Table III, and illustrated in Fig. 3. This geometry exemplifies a multilayer coil, well suited to test the leakage and coil resistance models. Utilizing (13) and (24), the total flux linkage of this device is plotted in Fig. 4, along with the numerical results from finite-element analysis (FEA) utilizing a 2-D and a 3-D model implemented in COMSOL. As can be observed, 13 simulations are performed until saturation. The accuracy of the magnetic model proposed in this section is comparable to the 3-D FEA. Using an Intel Core i7-8700 CPU

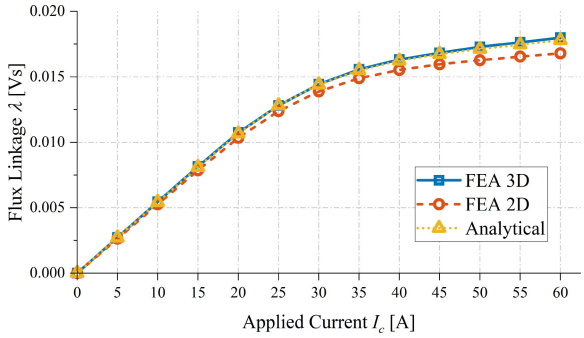


Fig. 4. Validation of the magnetic model (λ–*i* curve) using COMSOL FEA.

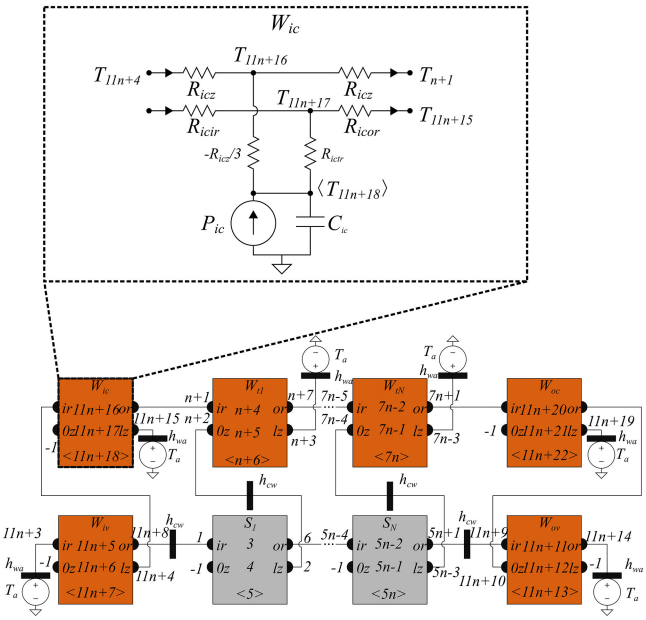


Fig. 5. TEC for the toroidal inductor.

@ 3.20 GHz with 16 GB of RAM computer, the 3-D FEA-based model takes 1 h 51 min to complete the 13 operating points' simulation. The 2-D FEA-based model takes 11 s, whereas the proposed analytical model described in this section needs on average 2.77 ms, using MATLAB 2018b.

C. Coil Electric Resistance

The ohmic losses can be derived from the geometry of the coil. The corresponding resistance is efficiently computed as

$$R_{\rm dc} = \frac{V_{cl}N^2}{k_{pf}A_{cl}^2\sigma\,(T)} \quad (25)$$

where N is the number of turns, $\sigma(T)$ is the conductivity of copper at a certain temperature T, and k_{pf} is the packing factor written as

$$k_{pf} = A_{cd}/A_{cl} \quad (26)$$

where the numerator is the conductor cross section A_{cd} defined as Na_c, hence constant around the coil. However, the coil cross-sectional area A_{cl} and volume V_{cl} are nonuniform in a

multilayer coil toroidal geometry. The coil is uniform throughout the vertical inner and outer sections; therefore, A_{cli} and A_{clo} are

$$A_{cli} = \pi d_{wi} \left(r_{ei} + r_a \right) \tag{27}$$

$$A_{clo} = \pi d_{wo} \left(r_{wo} + r_o \right). \tag{28}$$

In order to compute an accurate estimate of the coil resistance in the nonuniform coil sections, the equivalent coil cross-sectional area may be calculated by integrating the incremental resistance $dR = dl/(\sigma A(l))$, and extracting the equivalent area expression A_{eq} by equating the result to $R = l/(\sigma A_{eq})$. Thus, the coil cross section of the inner corner section A_{clic} can be approximated as

$$A_{clic} = \pi \frac{(d_{wi} + h_{ci})}{2} \sqrt{(2r_{ci})^2 - (2d_{ep} + (d_{wi} + h_{ci})/2)^2} \tag{29}$$

$$A_{cloc} = \pi \frac{(d_{wo} + h_{co})}{2} \sqrt{(2r_{co})^2 - (2d_{ep} + (d_{wo} + h_{co})/2)^2}. \tag{30}$$

As in (23), it is useful to divide the horizontal region of the coil into sections for integration with the heat transfer model. Each horizontal section coil cross-sectional area is

$$A_{cl,i} = 2\pi \left(h_{cs,i} r_{cs,i+1} - h_{cs,i+1} r_{cs,i} \right) / \ln \left(\frac{h_{cs,i} r_{cs,i+1}}{h_{cs,i+1} r_{cs,i}} \right). \tag{31}$$

Computing the total coil resistance using (25)–(31) for the inductor described in Table III yields 83.031 mΩ, computed in 59 μs, on average. Using the homogenized multiturn coil geometry analysis in a 3-D model in COMSOL, the dc resistance estimate 82.367 mΩ, computed in 2 min 28 s. The packing factor in each coil section can be estimated through (26). The coil volume V_{cl} is readily computed through the geometry, and the conductor volume is then estimated by $V_{cd} = k_{pf} V_{cl}$. From this quantity, the total wire length is estimated by $l_{wr} = V_{cd}/a_c$, which is constrained for manufacturability by the maximum length allowed in the winding machine header. Using the approach set forth in [28], the ac resistance of the coil resulting from skin effect can be estimated as follows:

$$R_{ac} = \frac{1}{n_{\text{spc}}} \text{real} \left\{ - \frac{V_{cd} J_0}{2\pi r_s a_c \sigma (T) \kappa J_0'} \right\} \tag{32}$$

where J_0 is the Bessel function of first kind of order zero, and κ is defined as

$$\kappa = \sqrt{\frac{j}{\omega \sigma (T) \mu}}. \tag{33}$$

Note that (32) yields one value for each harmonic in the current spectrum.

D. Proximity Losses

Finally, the proximity losses can be derived based on the magnetic energy stored around the coil, as further detailed in [28]. Assuming that these losses are linked only to the leakage flux around the wire, it can be derived that the proximity losses

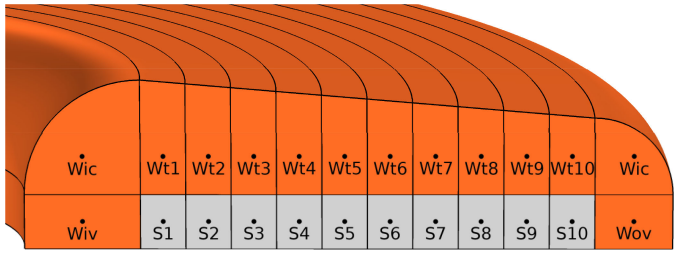


Fig. 6. Inductor described by Table III parameters, divided into ten sections.

are

$$P_{p,x} = \overline{\frac{di}{dt}}^2 \left(\frac{\mu_0 N^3 \pi \sigma (T_x) r_s^4 l_x P_{lk,x}}{4 V_{cl,x}} \right) \tag{34}$$

where the subscript "x" represents the section of the coil (inner vertical, outer vertical, corners, horizontal subsections, etc.), l_x is the length of such section, the bar indicates time average, and $P_{lk,x}$ comes from (18)–(23) for each specific region.

E. Core Losses

From the applied current waveform, the flux density waveform may be computed by solving (11) for a number of points in a symmetric period. For integration with the heat transfer model proposed in the next section, consider sampling B in the same number of sections of the core, such that it matches the divisions in the horizontal sections of the coil, as illustrated in Fig. 6. The power dissipated in each core section may be computed by any core losses formula of choice without any change to the rest of the inductor performance analysis [29]–[33]. In this work, the results are produced based on MSE (modified Steinmetz equation) introduced in [29].

V. Heat Transfer Model

In the inductor optimization, the device size is often limited not by electromagnetic performance, but rather thermal limits. This work includes a TEC model to couple with the electromagnetic losses previously described and predict the device temperature. Each element of the TEC was directly derived from the heat equation, hence accounting for temperature variation inside each region, despite the spatial average on each boundary, as derived in [34]. The TEC setup for this geometry is shown in Fig 5. Each box in the diagram corresponds to a cylindrical region in the TEC, as derived in [34]. Fig. 6 references each box in Fig. 5 to a specific region of the inductor, again the geometry described in Table III. Note that half-symmetry is used to reduce the system order. Since each cylindrical region contains a single uniformly distributed heat source, dividing the core into multiple sections allows the model to capture a discretized heat source across the core. This is used by the permeability optimization to modify how the losses are distributed across the core. Furthermore, notice that the corners and top sections of the coil are not cylindrical, as derived by the element used in the TEC. For the top sections of the core this is adjusted by assuming that the inner and outer heights are equal to their

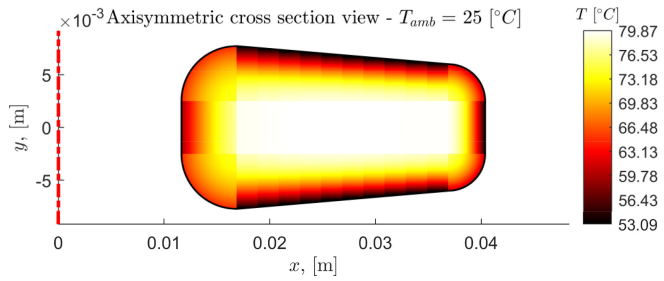


Fig. 7. Axisymmetric view of the approximate temperature distribution using the analytically derived TEC.

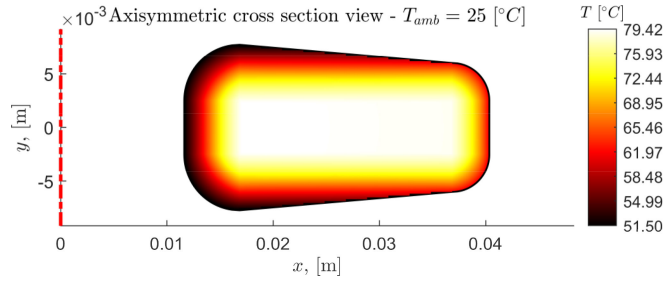


Fig. 8. Axisymmetric view of the temperature distribution using 3-D FEA.

average. The corners are sections of a torus. The outermost surface is exposed to the ambient temperature by a convective boundary condition with heat transfer coefficient h_{wa}. Consider bending this torus section into a cylinder with equivalent mean length, and exposed surface to the ambient. For the inner corner, the resulting equivalent cylindrical axial length is given by

$$l_{ice} = \frac{3\pi w_{mic} r_{ci} - 4w_{mic}^2}{6\pi r_{ci} - 12 w_{mic}} \tag{35}$$

where w_{mic} is defined by $(d_{wi} + h_{ci})/2$, and the inner corner equivalent inner and outer radii are

$$r_{icie} = \frac{l_{ice}}{w_{mic}} (\pi r_{ci} - 2w_{mic}) - \frac{w_{mic}^2}{4 l_{ice}} \tag{36}$$

$$r_{icoe} = \frac{l_{ice}}{w_{mic}} (\pi r_{ci} - 2w_{mic}) + \frac{w_{mic}^2}{4 l_{ice}}. \tag{37}$$

For the outer corner, the resulting equivalent cylindrical axial length is given by

$$l_{oce} = \frac{3\pi w_{moc} r_{co} + 4w_{moc}^2}{6\pi r_{co} + 12 w_{moc}} \tag{38}$$

where w_{moc} is defined by $(d_{wo} + h_{co})/2$, and the outer corner equivalent inner and outer radii are

$$r_{ocieq} = \frac{l_{oce}}{w_{moc}} (\pi r_{co} + 2w_{moc}) - \frac{w_{moc}^2}{4 l_{oce}} \tag{39}$$

$$r_{ocoe} = \frac{l_{oce}}{w_{moc}} (\pi r_{co} + 2w_{moc}) + \frac{w_{moc}^2}{4 l_{oce}}. \tag{40}$$

The protection layer is introduced in the TEC through a contact resistance h_{cw}, which has the equivalent

$$h_{cw} = \frac{4 h_{pl} k_a}{4k_a + h_{pl}(4 - \pi) r_s} \tag{41}$$

where h_{pl} is the protection layer heat transfer coefficient and k_a is the thermal conductivity of air. In order to reduce the complexity of the coil heat transfer analysis, the material properties are homogenized as a single solid with equivalent properties using the approach detailed in [34]. Note that both the core and homogenized coil sections are anisotropic. Once, each section property is computed, those are appropriately assigned to each orthogonal direction. Figs. 7 and 8 compare the results of the TEC proposed in this section against the 3-D FEA-based model implemented in COMSOL, using the geometry from Table III. Note that the proposed model captures the temperature distribution reasonably well and an accurate estimate of the peak temperature. The former is important to adjust the conductivity of the conductors in each region, until the losses converge, whereas the latter is used to constraint the peak temperature of the device. The semianalytical heat transfer model with ten sections takes on average 5.9 ms to simulate, whereas the 3-D FEA model takes 11 s. Finally, the electromagnetic and heat transfer models are coupled such that the coil temperature dependent losses from (32) and (34) are updated according to the mean temperature of the respective region. This is implemented in a fixed-point iteration algorithm until the temperature converges, or the maximum number of iterations is reached.

VI. Optimization Paradigm and Results

A. Inductor Optimization Paradigm

The design space for the permeability tuned toroidal inductor may be expressed as

$$\theta_L = [N \quad a_c^* \quad r_a \quad l_c \quad d_c \quad k_1 \quad \dots \quad k_n] \tag{42}$$

as defined in (5) as the independent geometric parameters, in addition to $k_1, \dots, k_n$, which are permeability multipliers as defined in (14). The number of multipliers to be optimized is defined by the user. In the fitness function, the geometry of the devices is built, from which the device mass M_L is computed. Subsequently, the electromagnetic performance is analyzed. The incremental inductance L_{inc} at the desired operating point is verified against the required value. From the core flux density, the core loss is computed using the method of choice, MSE, in the results presented in this article. Finally, the heat transfer analysis is computed with the coil losses until the temperature converges. The constraints in Table IV are checked as each corresponding variable is computed. If all constraints are satisfied, the fitness function is set to minimize mass and loss in the inductor

$$\min_{\theta_L} [P_L \quad M_L]^\top$$
$$\text{subject to} c_i = 1, \quad \forall i \tag{43}$$

where P_L is the total loss (core, proximity, ac winding losses) and c_i is 1 when the ith constraint in Table IV is satisfied.

In order to demonstrate the benefits of spatial permeability optimization, consider the design of a 0.75-mH inductor for a

TABLE IV
CONSTRAINTS IMPOSED TO THE OPTIMIZATION PROBLEM

Constraint	Expression	Value used	Remark
c_1	$l_{wr} \le l_{wrmx}$	$l_{wr} \le 9.5$ m	Maximum wire length
c_2	$k_{pf} \le k_{pfmx}$	$k_{pf} \le 0.7$	Maximum packing factor
c_3	$a_L \le a_{Lmx}$	$a_L \le 10$	Maximum aspect ratio
c_4	$M_L \le M_{Lmx}$	$M_L \le 0.5$ kg	Maximum weight
c_5	$L_{inc} \ge L_{req}$	$L_{inc} \ge 0.75$ mH	Minimum incremental inductance
c_6	$c_6 = 1$	$c_6 = 1$	TEC convergence flag
c_7	$P_l \le P_{mx}$	$P_l \le 100$ W	Maximum power loss
c_8	$T_{wpk} \le T_{wmx}$	$T_{wpk} \le 130$°C	Maximum winding temperature (class B)

TABLE V
DESIGN AND MODEL PARAMETERS

Variable	Value	Remark
I_{pk}	15 A	Sinusoidal waveform
f	1 kHz	Operating frequency
k_b	1.05	Winding built factor
d_{ep}	0.8 mm	Depth of protection layer
T_a	40 °C	Ambient Temperature
μ_{rmx}	100	Maximum achievable permeability
AWG_{mx}	18	Maximum strand wire gauge
N_s	10	Number of core sections
n_{spc}	1	Number of strands per conductor
h_{ca}	25 [W/(m²K)]	Heat transfer coef. winding to air
h_{pl}	560 [W/(m²K)]	Protection layer heat transfer coef.

15-A_{pk} current at 1 kHz, further detailed in Tables IV and V. For comparison purposes, multiobjective optimization is performed, allowing permeability tuning using three strategies, depending on the number of tuning parameters, as defined in (42). The first creates a constant permeability across the radius but adjusted for the application. The second creates an affine permeability profile, and the third a quadratic description. In this application and geometry, more tuning parameters do not provide any visible gain of performance in the Pareto optimal front. For comparison purposes, the optimization was also performed allowing only the relative magnetic permeability in commercial MPP powder toroidal cores as available in [35]. Rather than considering the geometries available in [35], this study allows the optimization of the dimensions, but the available relative permeabilities are constrained to the finite set in [35]: MPP14, MPP26, MPP60, MPP125, MPP147, MPP160, MPP173, MPP200, MPP300, and MPP500. The core loss characteristics of each core is according to [35]. The thermal conductivities of these cores are $k_x = k_y = k_z = 1.5\,[\mathrm{W}/(\mathrm{m}\cdot\mathrm{K})]$.

The Pareto optimal fronts are shown in Fig. 9. Notice therein that a significant gain is observed when the permeability is tuned for the application, and even further, when that is spatially tuned. In these conditions, the quadratic description of permeability provided a further reduction in losses in design solutions with mass between 0.225 and 0.45 kg. The permeability profiles of designs on the Pareto optimal front with three tuning parameters

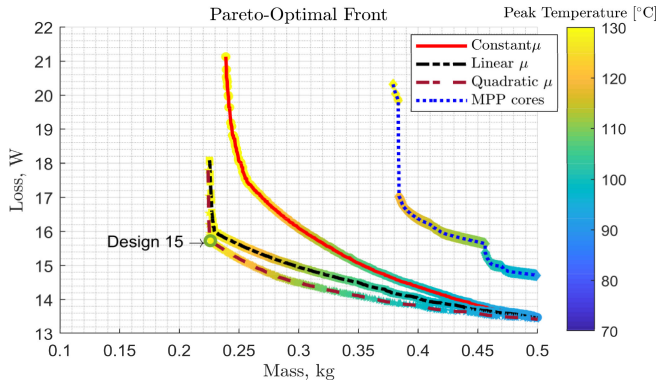


Fig. 9. Pareto optimal fronts and peak temperature of each design.

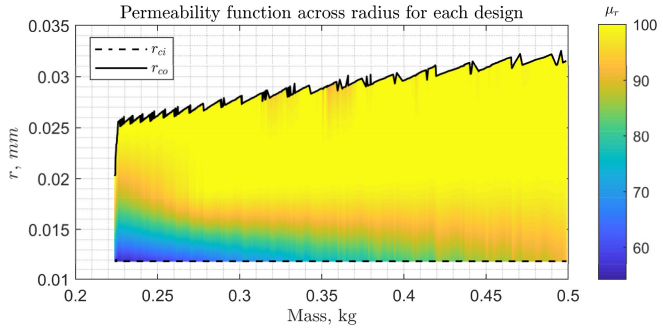


Fig. 10. Permeability profile for every design in the Pareto optimal front for the optimization results with three tuning parameters (quadratic profile).

(quadratic) are shown in Fig. 10. Notice that for smaller and less-efficient solutions, the optimization tends to maximize core utilization by spatially adjusting the permeability for a more even flux density distribution. In heavier and more efficient designs, the optimization tends to maximize the permeability (this is limited to 100 for the specific alloy used in this study) and flux density to meet the inductance requirement with fewer turns (reducing winding losses). This leads to similar performance for all tuned Pareto fronts above 0.45 kg.

In order to illustrate the features enabled by the simultaneous optimization of geometry and spatial permeability profile, consider Design 15 from the Pareto optimal front with quadratic order permeability (three tuning parameters) from Fig. 9. The flux density and temperature profiles of this inductor are depicted in Figs. 11 and 12, respectively. The former depicts the top view of the inductor, detailing the flux density inside of the core. The latter shows the cross-sectional temperature profile of the same design. Notice in Fig. 11 that the permeability profile was such that the flux density was nearly uniform and nearly saturated (see Fig. 1) except near the outer edge. At the outer edge, the optimized solution has a reduced flux density to decrease core losses in the outer region where the temperature reaches the maximum allowed value, as shown in Fig. 12.

B. System-Level Converter Optimization

Next, consider the benefits of spatial permeability manipulation in the scope of a power electronic converter optimization. Notice that, due to the computational efficiency of the

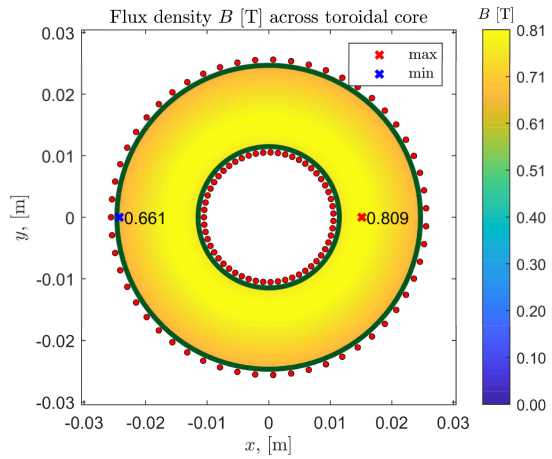


Fig. 11. Flux density (top view) of Design 15 from quadratic Pareto.

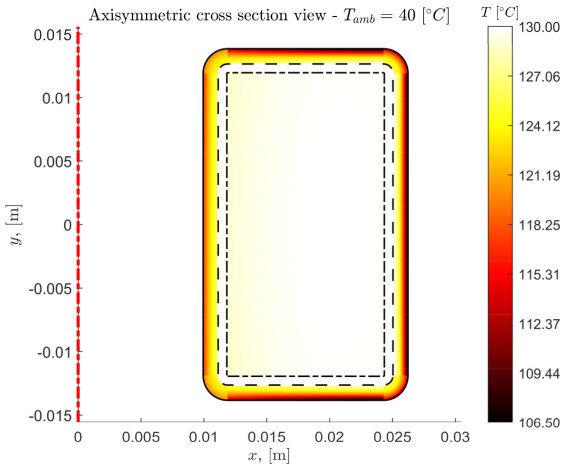


Fig. 12. Temperature profile of Design 15 from quadratic Pareto.

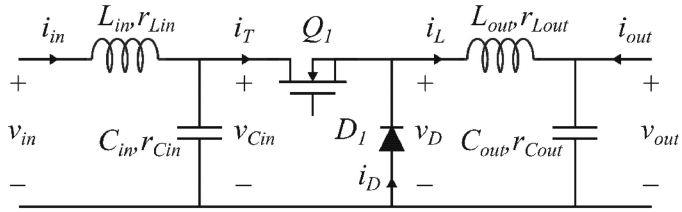


Fig. 13. Buck converter with input LC filter.

inductor optimization paradigm introduced in this work, the co-optimization of the components and independent converter operational parameters is attainable. In this instance, consider a buck converter with an input LC filter for conducted EMI (electromagnetic interference) mitigation, as depicted in Fig. 13. The optimization variables are

$$\theta_c = [f_{sw} \quad C_{\text{in}} \quad C_{\text{out}} \quad L_{\text{in}} \quad L_{\text{out}} \quad \theta_{Lin} \quad \theta_{Lout}] \tag{44}$$

where f_{sw} is the switching frequency, C_{in} and C_{out} are the input and output capacitance values, respectively, followed by their respective inductance values L_{in} and L_{out}. θ_{Lin} and θ_{Lout} are the independent geometric and permeability tuning parameters defined in (42) for the input and output inductors, respectively. In similar analysis implemented in [36], from average value steady-state analysis, and enforcing continuous conduction mode, the ripple input capacitor voltage $\Delta v_{C\text{in}}$ is

$$\Delta v_{C\text{in}} = \bar{i}_L d\,(1-d)\,/\,(C_{\text{in}} f_{sw}) \tag{45}$$

where $\bar{i}_L$ is the average output inductor defined as $-\bar{i}_{\text{out}} = P_{\text{out}}/v_{\text{out}}$ from Fig. 13, and d is the duty cycle. The input inductor current ripple Δi_{in} is

$$\Delta i_{\text{in}} = \Delta v_{C\text{in}} / \left(8 L_{\text{in}} f_{sw}\right). \tag{46}$$

The output capacitor voltage ripple is

$$\Delta v_{\text{out}} = \Delta i_L / \left(8 C_{\text{out}} f_{sw}\right) \tag{47}$$

where the output inductor current ripple Δi_L is

$$\Delta i_L = \left(\bar{v}_{\text{in}} - r_{L\text{in}} \bar{i}_T - v_{fsw} + v_{fd}\right) d\,(1-d)\,/\,(f_{sw} L_{\text{out}}) \tag{48}$$

where $\bar{i}_T$ is the average transistor current defined as $d\bar{i}_L$, r_{Lin} is the input inductor resistance computed using (25)–(31) at the final operating temperature (which can vary significantly from the resistance at ambient temperature), v_{fsw} is the MOSFET forward voltage drop, and v_{fd} is the diode forward voltage drop. The semiconductor voltage drops are determined from each component loss

$$v_{fsw} = \left(P_{tcd} + P_{tsw}\right) / \left(d\bar{i}_L\right) \tag{49}$$

where P_{tcd} and P_{tsw} are conduction and switching losses, respectively. And v_{fd} is

$$v_{fd} = P_d / \left((1-d)\,\bar{i}_L\right) \tag{50}$$

where P_d is the diode conduction losses (SiC majority carrier diodes are selected where reverse recovery current and corresponding switching losses are negligible). In this article, the MOSFET Q_1 used is Cree C2M0080120D, and diode D_1 is Cree C4D20120A. The semiconductor losses, and heat sink and capacitor mass are computed as described in [36]. The steady-state duty cycle is computed as

$$d = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \tag{51}$$

where

$$a = -r_{L\text{in}} \bar{i}_L \tag{52}$$

$$b = \bar{v}_{\text{in}} - v_{fsw} + v_{fd} \tag{53}$$

$$c = -v_{fd} - \bar{v}_{\text{out}} - r_{L\text{out}} \bar{i}_L. \tag{54}$$

Notice that not only (49)–(51) are coupled, but also both inductor resistances (r_{Lin} and r_{Lout}) are temperature dependent. Thus, the inductor losses couple the resistances back to duty cycle and current ripple. The nonlinear system of equations can be solved numerically using a fixed-point iteration, as depicted in Algorithm 1 in Table VI.

TABLE VI
ALGORITHM FOR BUCK CONVERTER STEADY-STATE ANALYSIS

Algorithm 1: Buck Converter Steady State Analysis

Input: Converter Parameters in (44) and Table VII.

Output: $\overline{i_{in}}$, $\overline{i_L}$, Δi_L, d, v_{fd}, v_{fsw}, P_{tcd}, P_{tsw}, P_d, r_{Lout},P_{Lout}, M_{Lout}, r_{Lint}, P_{Lin}, M_{Lin}.

1: Initialize $\overline{i_{in}}$, $\overline{i_L}$, d, v_{fd}, v_{fsw}, and Δi_L assuming 100% efficiency.
2: **while** $k \le k_{max}$ AND $error \ge error_{max}$ **do**
3: Create i_L triangular waveform based on $\overline{i_L}$, d, and Δi_L.
4: Extract N_h largest harmonics from i_L waveform for winding losses
5: Evaluate output inductor using (1)-(42), return r_{Lout} at converged operating temperature, losses P_{Lout} and mass M_{Lout}.
6: **if** any $c_i \ne 1,\ \forall i \in [1,8]$ **then**
7: **return 0**
8: **end if**
9: Compute Δi_{in} using (45)-(46), and create i_{in} triangular waveform
11: Extract N_h largest harmonics from i_{in} waveform for winding losses
12: Evaluate input inductor using (1)-(42), return r_{Lin} at converged operating temperature, losses P_{Lin} and mass M_{Lin}.
13: **if** any $c_i \ne 1,\ \forall i \in [1,8]$ **then**
14: **return 0**
15: **end if**
16: Update Δi_L and d using (48) and (51).
17: **if** $(\overline{i_L} - \Delta i_L/2) \le 0$ OR $d \notin [0,1]$ **then**
18: **return 0**
19: **end if**
20: Update $\overline{i_{in}}$, P_{tcd}, P_{tsw}, P_d, v_{fsw}, and v_{fd}.
21: Update $error = \sqrt{\left(v_{fsw}^{new} - v_{fsw}^{old}\right)^2 + \left(v_{fd}^{new} - v_{fd}^{old}\right)^2}$, and increment k.
22: **end while**
23: **if** $error \ge error_{max}$ **then**
24: **return 0**
25: **else**
26: **return** $\overline{i_{in}}$, $\overline{i_L}$, Δi_L, d, v_{fd}, v_{fsw}, P_{tcd}, P_{tsw}, P_d, r_{Lout}, P_{Lout}, M_{Lout}, r_{Lint}, P_{Lin}, M_{Lin}.
27: **end if**

From the converged analysis, the total losses P considered are accumulated

$$P = P_{\mathrm{Lin}} + P_{\mathrm{Lout}} + P_d + P_{tsw} + P_{tcd} \tag{55}$$

where P_{Lin} and P_{Lout} are the input and output inductor losses, respectively. And the total mass is defined as

$$M = M_{\mathrm{Lin}} + M_{\mathrm{Lout}} + M_{HS} + M_{\mathrm{Cin}} + M_{\mathrm{Cout}} \tag{56}$$

where M_{HS} is the mass of the heat sink, and M_{Cin} and M_{Cout} are, respectively, the input and output capacitor masses computed as in [36]. The design specifications in Table V are applied to each inductor evaluation, with exception of the excitation current waveform that is defined by the converter parameters. The optimization is set up to minimize both (55) and (56) while meeting the converter ripple constraints defined in Table VII, which are computed using (45)–(48). Consider the illustrative example of an 800–400-V buck converter, optimized to operate at 10 kW. The resultant Pareto optimal fronts are depicted in Fig. 14. Similar to Section VI-A, the previously mentioned selection of permeabilities in [35] are made available for the inductor geometry optimization to demonstrate optimal results when the permeability cannot be tuned for the application. A significant improvement is provided by the permeability optimization, especially when spatially manipulated. The inductor optimization

TABLE VII
EXTRA CONSTRAINTS IMPOSED TO THE CONVERTER OPTIMIZATION PROBLEM

	Expression	Remark
c_a	$\Delta v_{out} / \overline{v_{out}} \le 1\%$	Peak-to-peak output capacitor voltage ripple
c_b	$\Delta i_L / \overline{i_L} \le 20\%$	Peak-to-peak output inductor current ripple
c_c	$\Delta v_c / \overline{v_c} \le 5\%$	Peak-to-peak input capacitor voltage ripple
c_d	$\Delta i_{in} / \overline{i_{in}} \le 1\%$	Peak-to-peak input inductor current ripple

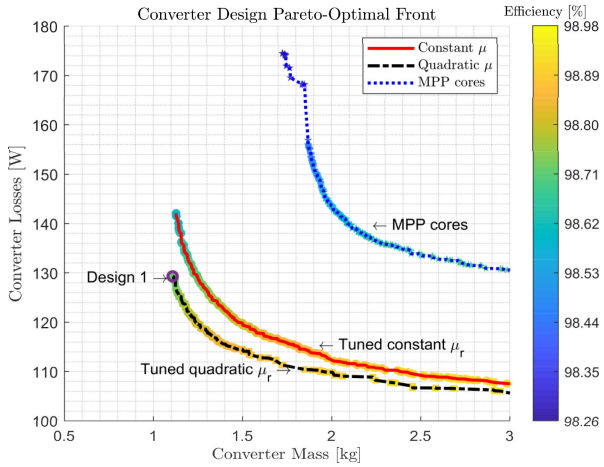


Fig. 14. Converter Pareto optimal fronts.

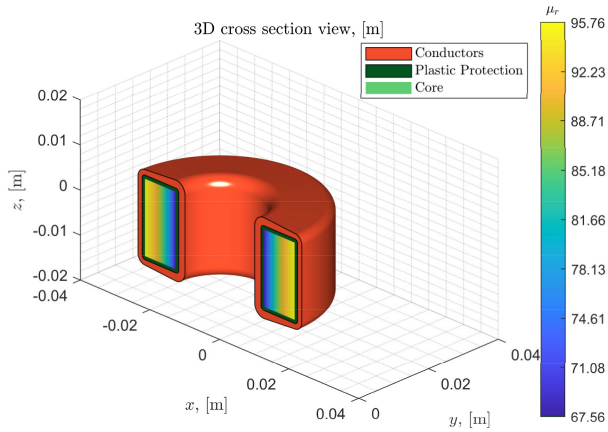


Fig. 15. Input inductor of Design 1 with quadratic spatial permeability.

with four permeability multipliers or more did not provided further performance gains. Consider the converter design solution 1 depicted in Fig. 14. The optimized variables in (44) in this design selected the switching frequency at 61.727 kHz and 144.3 and 109.9 μF for input and output capacitance, respectively. The inductor geometries and permeability profiles of Design 1 are depicted in Figs. 15 and 16, which were optimized for 496.9 and 645.2 μH for input and output inductances, respectively.

VII. CONCLUSION

In this article, the benefits of spatial permeability optimization are evaluated through the comparison of Pareto optimal fronts, in

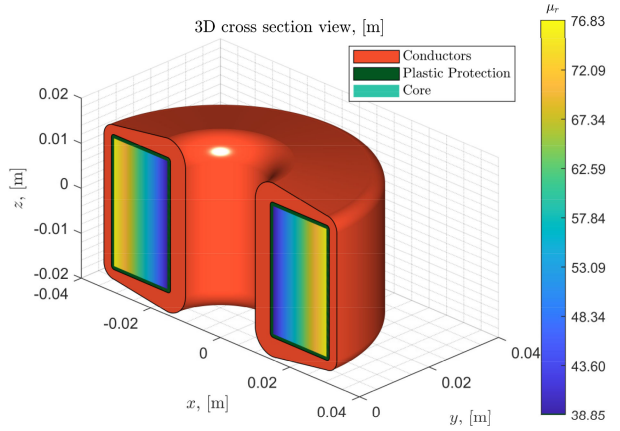


Fig. 16. Output inductor of Design 1 with quadratic spatial permeability.

the context of both inductor- and converter-level optimization. Thus, the introduction of a new design variable is evaluated against traditional alternatives in an optimization paradigm. It can be observed that the spatial permeability tuning offers a superior performance that cannot be compensated by geometric optimization of the topology herein considered. Furthermore, a computationally efficient multiphysics model for toroidal multilayer inductor analysis was introduced. The software implemented in MATLAB 2018b is available online as an attachment to this article [37].

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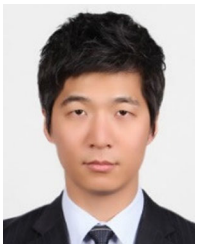
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